gravitational waves; Figure 7.4), it is left totally "hairless": *All* of its properties are precisely determined by just three numbers, its mass, its angular momentum, and its electric charge. The hole has no randomness whatsoever.

Box 12.3

Entropy in a Child's Playroom

Imagine a square playroom containing 20 toys. The floor of the room is made of 100 large tiles (with 10 tiles running along each side), and a father has cleaned the room, throwing all the toys onto the northernmost row of tiles. The father cared not one whit which toys landed on which tiles, so they are all randomly scrambled. One measure of their randomness is the number of ways that they could have landed (each of which the father considers as equally satisfactory), that is, the number of ways that the 20 toys can be distributed over the 10 tiles of the northern row. This number turns out to be $10 \times 10 \times 10 \times ... \times 10$, with one factor of 10 for each toy; that is, 10^{20}.

This number, 10^{20}, is one description of the amount of randomness in the toys. However, it is a rather unwieldly description, since 10^{20} is such a big number. More easy to manipulate is the *logarithm* of 10^{20}, that is, the number of factors of 10 that must be multiplied together to get 10^{20}. The logarithm is 20; and *this logarithm of the number of ways the toys could be scattered over the tiles is the toys' entropy.*

Now suppose that a child comes into the room and plays with the toys, throwing them around with abandon, and then leaves. The father returns and sees a mess. The toys are now far more randomly distributed than before. Their entropy has increased. The father doesn't care just where each toy is; all he cares is that they have been scattered randomly throughout the room. How many different ways might they have been scattered? How many ways could the 20 toys be distributed over the 100 tiles? $100 \times 100 \times 100 \times ... \times 100$, with one factor of 100 for each toy; that is, $100^{20} = 10^{40}$ ways. The logarithm of this number is 40, so the child increased the toys' entropy from 20 to 40.